

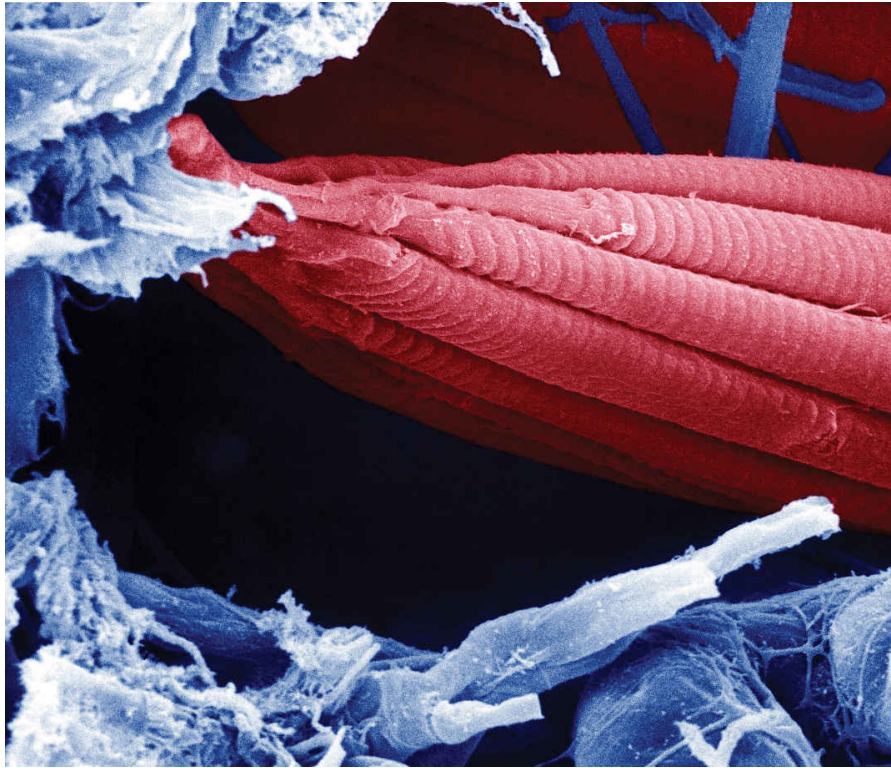
PART I

Exercising Muscle

In the introduction, we explored the foundations of exercise and sport physiology. We defined these fields of study, gained a historical perspective of their development, looked at present trends as well as the future of exercise physiology, and established some basic concepts that we will follow through the remainder of this book. We also examined the tools and research methods used by exercise physiologists along with some tips about interpreting graphs and scientific notation. With this foundation, we can begin our main objective—understanding how the human body performs, and adapts to, exercise and physical activity. Because muscle is the foundation of all movement, we start with [chapter 1](#), Structure and Function of Exercising Muscle, where we focus on skeletal muscle, examining the structure and function of skeletal muscles and muscle fibers and how they contract. We learn how muscle fiber types differ and why these differences are important to specific types of activity. Because movement requires energy, in [chapter 2](#), Fuel for Exercise: Bioenergetics and Muscle Metabolism, we study the principles of metabolism, focusing on the primary form of usable energy, adenosine triphosphate (ATP), and how it is provided from the foods that we eat through three energy systems. In [chapter 3](#), Neural Control of Exercising Muscle, we discuss how the nervous system initiates and controls muscle actions. [Chapter 4](#), Hormonal Control During Exercise, presents an overview of the complex endocrine system, with a focus on hormonal control of energy metabolism, body fluid and electrolyte balance during exercise, and caloric intake. Finally, [chapter 5](#), Energy Expenditure, Fatigue, and Muscle

Soreness, discusses the measurement of energy expenditure, how energy expenditure changes from rest to varying types and intensities of exercise, the various causes of fatigue that limits exercise performance, and the causes of muscle cramping and feelings of soreness.





1

Structure and Function of Exercising Muscle

In this chapter and in the web study guide

Anatomy of Skeletal Muscle

Muscle Fibers

Myofibrils

AUDIO FOR FIGURE 1.2 describes the structures of a muscle.

ACTIVITY 1.1 *Muscle Structure* reviews the basic structures of muscle.

AUDIO FOR FIGURE 1.3 describes the structure of a muscle fiber.

ACTIVITY 1.2 *Structure of a Skeletal Muscle Cell* reviews the structures in a single muscle fiber.

AUDIO FOR FIGURE 1.5 describes the structure of a sarcomere.

ACTIVITY 1.3 *Structure of the Sarcomere* reviews the structures in a sarcomere.

Muscle Fiber Contraction

Excitation–Contraction Coupling

Role of Calcium in the Muscle Fiber

The Sliding Filament Theory: How Muscles Create Movement

Energy for Muscle Contraction

Muscle Relaxation

AUDIO FOR FIGURE 1.7 describes the structure of a motor unit.

ANIMATION FOR FIGURE 1.8 breaks down excitation–contraction coupling.

ACTIVITY 1.4 *Sliding Filament Theory* describes this theory of muscle contraction and explores what happens at the cellular and gross motor movement levels.

ANIMATION FOR FIGURE 1.9 shows the function of a sarcomere during muscle contraction.

ANIMATION FOR FIGURE 1.10 shows the steps of the contractile cycle in a sarcomere.

Muscle Fiber Types

Characteristics of Type I and Type II Fibers

Distribution of Fiber Types

Fiber Type and Exercise

Determination of Fiber Type

AUDIO FOR FIGURE 1.11 describes the muscle fiber distinctions.

ACTIVITY 1.5 *Fiber Types* differentiates between type I and type II skeletal muscle fibers.

Skeletal Muscle and Exercise

Muscle Fiber Recruitment

Fiber Type and Athletic Success

Muscle Contraction

ACTIVITY 1.6 *Fiber Recruitment* tests your understanding of the types of muscle fibers recruited and the order of recruitment.

ACTIVITY 1.7 *Generation of Force* reviews the factors that influence the development of muscle force.

AUDIO FOR FIGURE 1.14 describes the concepts of twitch, summation, and tetanus.

AUDIO FOR FIGURE 1.15 describes the variation in force with changes in sarcomere length.

AUDIO FOR FIGURE 1.16 explains the relationship between velocity and muscle force production.

In Closing

Liam Hoekstra possesses a physique and physical attributes like many professional athletes: ripped abdominal muscles, enough strength to perform feats like an iron cross and inverted sit-ups, and amazing speed and agility. Not bad when you consider the fact that Liam could do all this when he was just 19 months old and weighed 10 kg (22 lb)! Liam has a rare genetic condition called myostatin-related muscle hypertrophy, a condition that was first described in an abnormally muscular breed of beef cattle in the late 1990s. Myostatin is a protein that inhibits the growth of skeletal muscles; myostatin-related muscle hypertrophy is a genetic mutation that blocks production of this inhibitory growth factor and thus promotes the rapid growth and development of skeletal muscles.

Liam's condition is extremely rare in humans, with fewer than 100 cases documented worldwide. However, studying this genetic phenomenon has helped scientists unlock secrets of how skeletal muscles grow and deteriorate. Research on Liam's condition could lead to new treatments for debilitating muscular conditions such as muscular dystrophy. On the darker side, it could open up a whole new realm of abuse by athletes who are looking for shortcuts to develop muscle size and strength, not unlike the illicit and dangerous use of anabolic steroids.

When the heart beats, when partially digested food moves through the intestines, and when the body moves in any way, muscle is involved. These many and varied functions of the muscular system are performed by three distinct types of muscle (see [figure 1.1](#)): smooth muscle, cardiac muscle, and skeletal muscle.

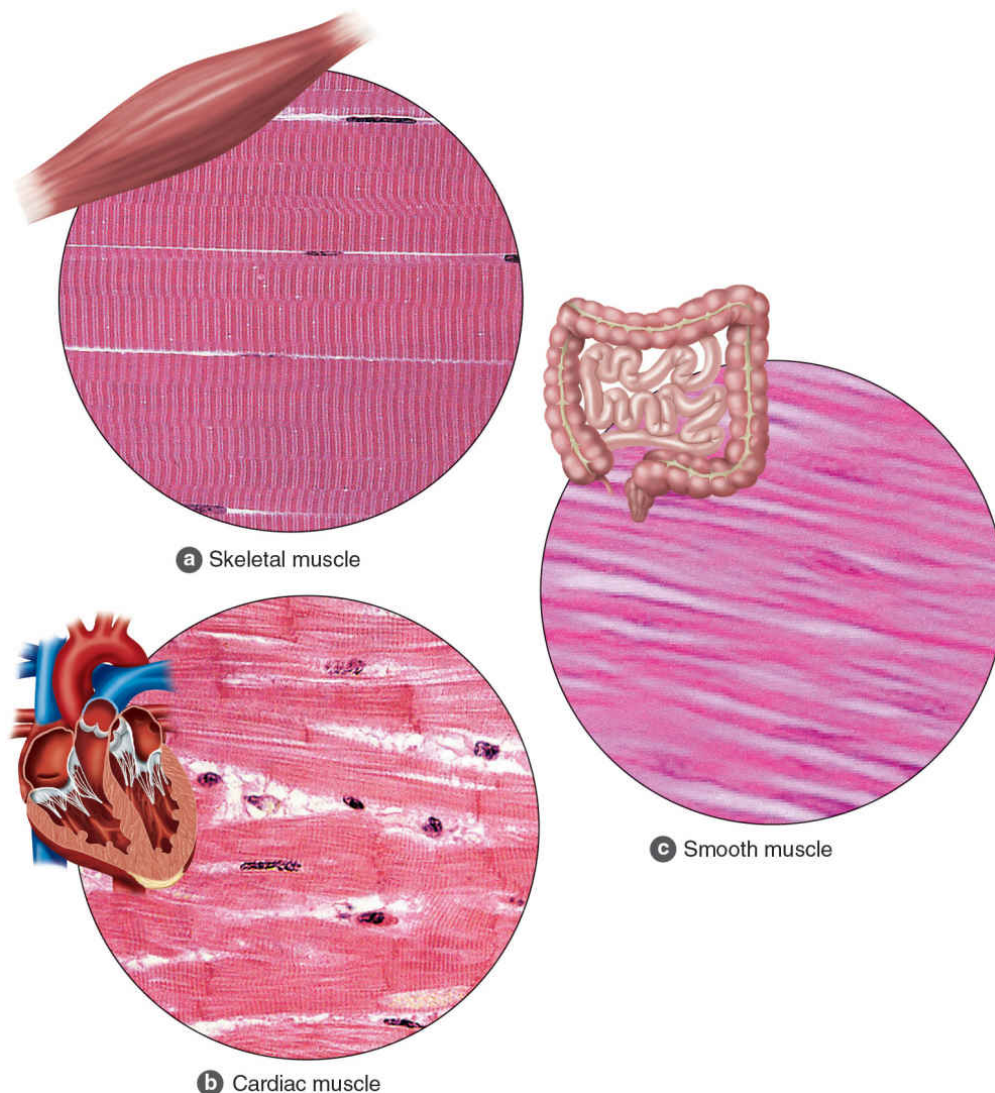


FIGURE 1.1 Microscopic photographs of the three types of muscle: (a) skeletal, (b) cardiac, and (c) smooth.

Smooth muscle is sometimes called involuntary muscle because it is not under direct conscious control. It is found in the walls of most blood vessels, where its contraction or relaxation leads to vessel constriction or dilation, respectively, to regulate blood flow. It is also found in the walls of most internal organs, allowing them to contract and relax, for example, to move food through the digestive tract, to expel urine, or to give birth.

Cardiac muscle is found only in the heart, composing the vast majority of the heart's structure. It shares some characteristics with skeletal muscle, but like smooth muscle, it is not under conscious control. Cardiac muscle in essence controls itself, with some fine-

tuning by the nervous and endocrine systems. Cardiac muscle is discussed more fully in [chapter 6](#).

Skeletal muscles are under conscious control and are so named because most attach to and move the skeleton. Together with the bones of the skeleton, they make up the **musculoskeletal system**. The names of many of these muscles have found their way into our everyday vocabulary—such as deltoids, pectorals (or “pecs”), and biceps—but the human body contains more than 600 skeletal muscles. The thumb alone is controlled by nine separate muscles!

Exercise requires movement of the body, which is accomplished through the action of skeletal muscles. Because exercise and sport physiology depend on human movement, the primary focus of this chapter is on the structure and function of skeletal muscle. Although the anatomical structures and control of smooth, cardiac, and skeletal muscle differ in many respects, their principles of action—for example, creating tension, shortening, and lengthening—are similar.

Anatomy of Skeletal Muscle

When we think of muscles, we visualize each muscle as a single unit. This is natural because a skeletal muscle most often acts as a single entity. But skeletal muscles are far more complex than that.

If a person were to dissect a muscle, he or she would first cut through an outer connective tissue covering known as the **epimysium** (see [figure 1.2](#)). It surrounds the entire muscle and functions to hold it together and give it shape. Once through the epimysium, one would see small bundles of fibers wrapped in a connective tissue sheath. These bundles are called fascicles (or fasciculi), and the connective tissue sheath surrounding each **fascicle** is the **perimysium**.

Finally, by cutting through the perimysium and using a microscope, one would see the individual **muscle fibers**, each of which is a muscle cell. Unlike most cells in the body, which have a single nucleus, muscle cells are multinucleated. A sheath of connective tissue, called the **endomysium**, also covers each muscle fiber. It is generally thought that muscle fibers extend from one end of the muscle to the other, but under the microscope, muscle bellies (the

thick middle parts of muscles) often divide into compartments or more transverse fibrous bands (inscriptions).

Because of this compartmentalization, the longest human muscle fibers are about 12 cm (4.7 in.), which corresponds to about 500,000 sarcomeres, the basic functional unit of the myofibril. The number of fibers in different muscles ranges from several hundred (e.g., in the tensor tympani, attached to the eardrum) to more than a million (e.g., in the medial gastrocnemius muscle).¹²

Muscle Fibers

Muscle fibers range in diameter from 10 to 120 μm , so they are nearly invisible to the naked eye. The following sections describe the structure of the individual muscle fiber.

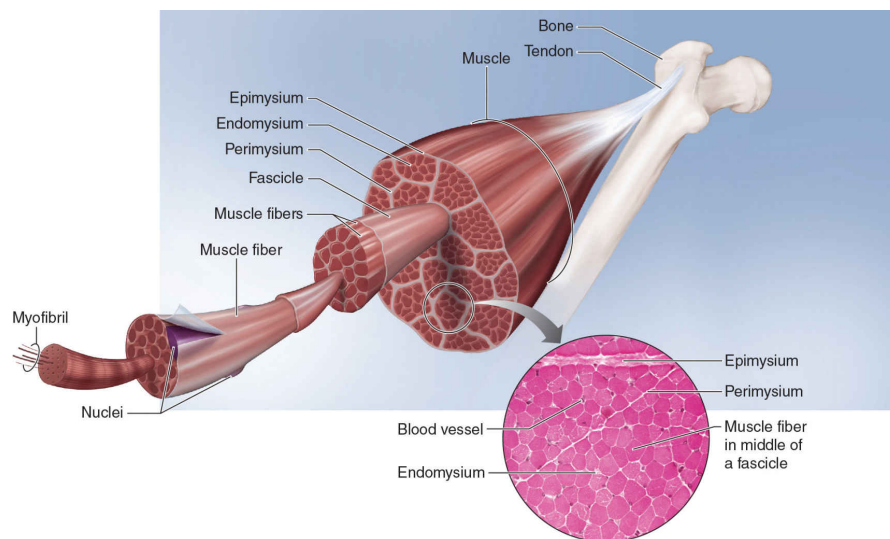


FIGURE 1.2 The basic structure of muscle.



Plasmalemma

Looking closely at an individual muscle fiber, it is surrounded by a plasma membrane, called the **plasmalemma** (figure 1.3). The plasmalemma is part of a larger unit referred to as the **sarcolemma**.

The sarcolemma is composed of the plasmalemma and the basement membrane. (Some textbooks use the term *sarcolemma* to refer to just the plasmalemma.¹²) At the end of each muscle fiber, its plasmalemma fuses with the tendon, which inserts into the bone. Tendons are made of fibrous cords of connective tissue that transmit the force generated by muscle fibers to the bones, thereby creating motion. So typically, individual muscle fibers are ultimately attached to bone via the tendon.

RESEARCH PERSPECTIVE 1.1

Muscle Changes After Only 6 Weeks of Training

Architectural characteristics of a muscle, such as its thickness, pennation angle (the angle at which the fibers are oriented within the muscle), and fascicle length, all contribute to its ability to produce force. Changes in structural characteristics have been demonstrated in many muscles in response to mechanical stimuli, including long-term exercise training. Understanding how—and how quickly—muscle architecture adapts to exercise training is important for recreational exercisers beginning a training program and athletes getting ready for competition.

A group of investigators recently used ultrasound imaging to examine the architectural adaptations of the biceps femoris in a group of young men before and after 6 weeks of either concentric or eccentric strength training.¹⁷ Eccentric strength training increased muscle fascicle length and reduced pennation angle (the fascicles aligned better with the direction of the muscle). In contrast, concentric strength training reduced fascicle length and increased pennation angle (fascicles were angled more away from the full muscle's direction). After 4 weeks of detraining, eccentric training-induced alterations were reversed, but the adaptations in response to concentric training were maintained. Thus, short-term resistance training can cause structural adaptations in the biceps that are highly specific to the mode of training. Understanding architectural alterations that occur in response to training is important for injury prevention and development of proper rehabilitation programs.

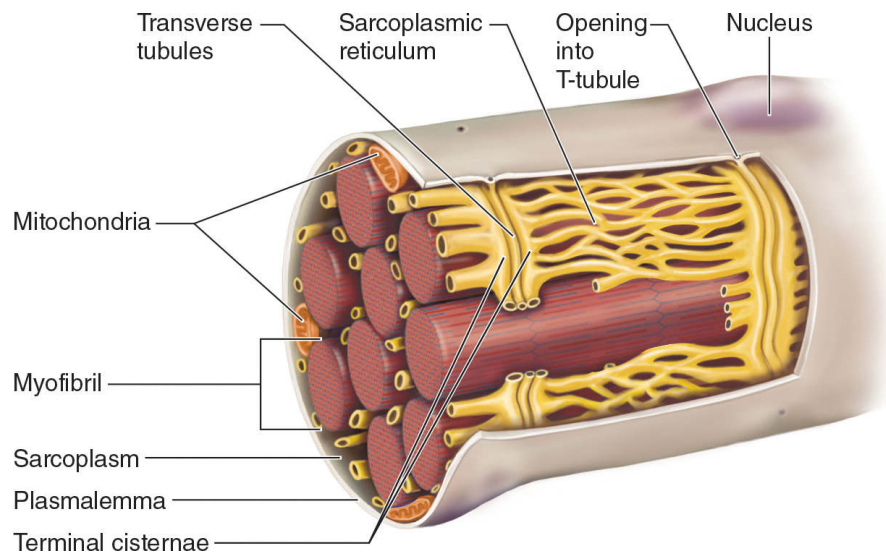


FIGURE 1.3 The structure of a single muscle fiber.



The plasmalemma has several unique features that are critical to muscle fiber function. It appears as a series of shallow folds along the surface of the fiber when the fiber is contracted or in a resting state, but these folds disappear when the fiber is stretched. This folding allows stretching of the muscle fiber without disrupting the plasmalemma. The plasmalemma also has junctional folds in the innervation zone at the motor end plate, which assists in the transmission of the action potential from the motor neuron to the muscle fiber, as discussed later in this chapter. Finally, the plasmalemma helps to maintain acid–base balance and transport of metabolites from the capillary blood into the muscle fiber.¹²

Satellite cells are located between the plasmalemma and the basement membrane. These cells are involved in the growth and development of skeletal muscle and in muscle's adaptation to injury, immobilization, and training. This is discussed in greater detail in subsequent chapters.

Sarcoplasm

Inside the plasmalemma, a muscle fiber contains successively smaller subunits, as shown in [figure 1.3](#). The largest of these are myofibrils, the contractile element of the muscle, which are described later. A gelatin-like substance fills the spaces within and between the myofibrils. This is the **sarcoplasm**. It is the fluid part of the muscle fiber—its cytoplasm. The sarcoplasm contains mainly dissolved proteins, minerals, glycogen, fats, and necessary organelles. It differs from the cytoplasm of most cells because it contains a large quantity of stored glycogen as well as the oxygen-binding compound myoglobin, which is similar in structure and function to the hemoglobin found in red blood cells.

Transverse Tubules The sarcoplasm also houses an extensive network of **transverse tubules (T-tubules)**, which are extensions of the plasmalemma that pass laterally through the muscle fiber. These tubules are interconnected as they pass among the myofibrils, allowing nerve impulses received by the plasmalemma to be transmitted rapidly to individual myofibrils. The tubules also provide pathways from outside the fiber to its interior, enabling substances to enter the cell and waste products to leave the fibers.

Sarcoplasmic Reticulum A longitudinal network of tubules, known as the **sarcoplasmic reticulum (SR)**, is also found within the muscle fiber. These membranous channels parallel the myofibrils and loop around them. The SR serves as a storage site for calcium, which is essential for muscle contraction. [Figure 1.3](#) depicts the T-tubules and the SR. Their functions are discussed in more detail later in this chapter when we consider the process of muscle contraction.

Myofibrils

Each muscle fiber contains several hundred to several thousand **myofibrils**. These small fibers are made up of the basic contractile elements of skeletal muscle—the sarcomeres. Under the electron microscope, myofibrils appear as long strands of sarcomeres.

Sarcomeres

Under a light microscope, skeletal muscle fibers have a distinctive striped appearance. Because of these markings, or striations, skeletal

muscle is also called striated muscle. This striation pattern is also seen in cardiac muscle, so it too can be considered striated muscle.

Refer to [figure 1.4](#), showing myofibrils within a single muscle fiber, and note the striations. Note that dark regions, known as A-bands, alternate with light regions, known as I-bands. Each dark A-band has a lighter region in its center, the H-zone, which is visible only when the myofibril is relaxed. There is a dark line in the middle of the H-zone called the M-line. The light I-bands are interrupted by a dark stripe referred to as the Z-disk, also known as the Z-line.

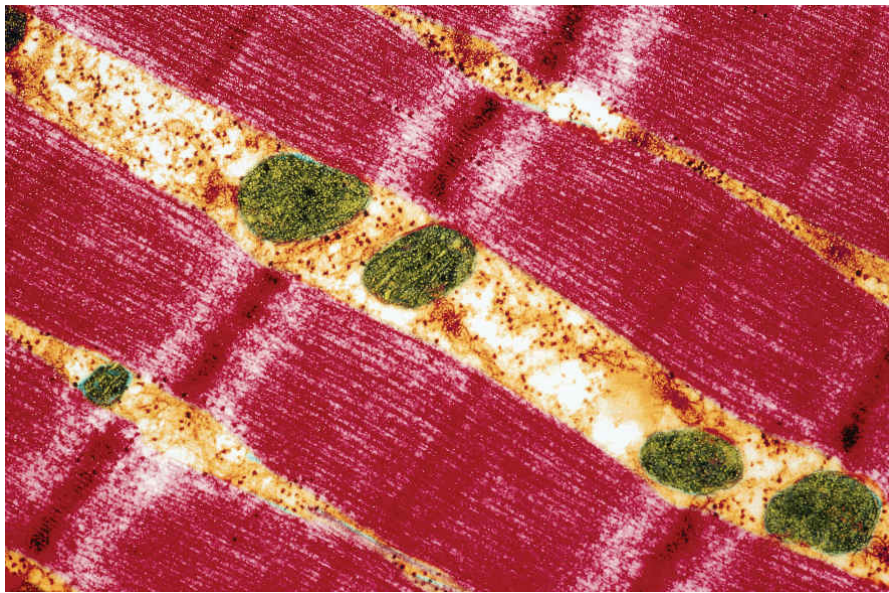


FIGURE 1.4 An electron micrograph of myofibrils within a muscle fiber showing mitochondria (green) between the myofibrils.

A **sarcomere** is the basic functional unit of a myofibril and the basic contractile unit of muscle. Each myofibril is composed of numerous sarcomeres joined end to end at the Z-disks. Each sarcomere includes several elements found between each pair of Z-disks, in this sequence:

- An I-band (light zone)
- An A-band (dark zone)
- An H-zone (in the middle of the A-band)
- An M-line in the middle of the H-zone
- The rest of the A-band
- A second I-band

In Review

- An individual muscle cell is called a muscle fiber.
- Muscle fibers have a cell membrane and the same organelles—mitochondria, lysosomes, and so on—as other cell types but are uniquely multinucleated.
- A muscle fiber is enclosed by a plasma membrane called the plasmalemma.
- The cytoplasm of a muscle fiber is called the sarcoplasm.
- The extensive tubule network found in the sarcoplasm includes T-tubules, which allow communication and transport of substances throughout the muscle fiber, and the SR, which stores calcium.
- The sarcomere is the smallest functional unit of a muscle.

Looking at individual myofibrils through an electron microscope, one can differentiate two types of small protein filaments that are responsible for muscle contraction. The thinner filaments are composed primarily of **actin**, and the thicker filaments are primarily **myosin**. The striations seen in muscle fibers result from the alignment of these filaments, as illustrated in [figure 1.4](#). The light I-band indicates the region of the sarcomere where there are only thin filaments. The dark A-band represents the regions that contain both thick and thin filaments. The H-zone is the central portion of the A-band and contains only thick filaments. The absence of thin filaments causes the H-zone to appear lighter than the adjacent A-band. In the center of the H-zone is the M-line, which is composed of proteins that serve as the attachment site for the thick filaments and assist in stabilizing the structure of the sarcomere. Z-disks, composed of proteins, appear at each end of the sarcomere. Along with two additional proteins, titin and nebulin, they provide points of attachment and stability for the thin filaments.

Thick Filaments

About two-thirds of all skeletal muscle protein is myosin, the principal protein of the thick filament. Each myosin filament typically contains about 200 myosin molecules.

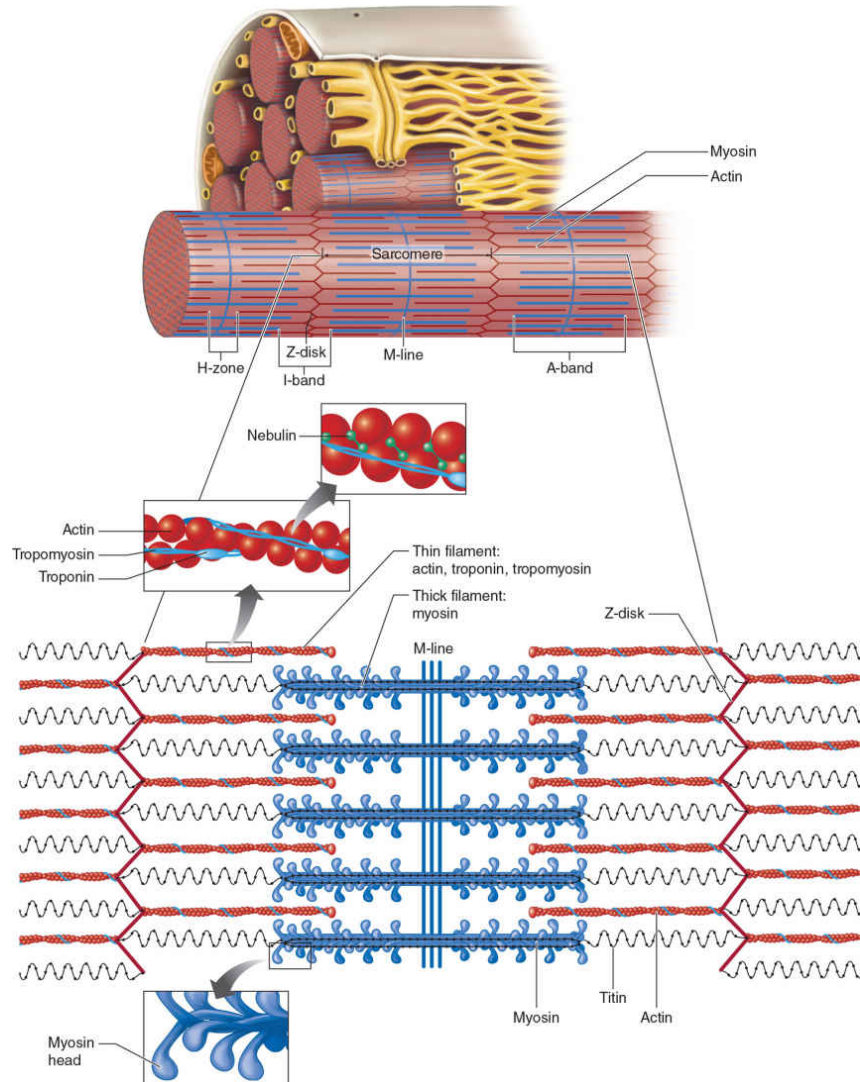


FIGURE 1.5 The sarcomere contains a specialized arrangement of actin (thin) and myosin (thick) filaments. The role of titin is to position the myosin filament to maintain equal spacing between the actin filaments. Nebulin is often referred to as an anchoring protein because it provides a framework that helps stabilize the position of actin.



Each myosin molecule is composed of two protein strands twisted together (see [figure 1.5](#)). One end of each strand is folded into a globular head, called the myosin head. Each thick filament contains many such heads, which protrude from the thick filament to form

cross-bridges that interact during muscle contraction with specialized active sites on the thin filaments. There is an array of fine filaments, composed of **titin**, that stabilize the myosin filaments along their longitudinal axis (see [figure 1.5](#)). Titin filaments extend from the Z-disk to the M-line.

Thin Filaments

Each thin filament, although often referred to simply as an actin filament, is actually composed of three different protein molecules—actin, **tropomyosin**, and **troponin**. Each thin filament has one end inserted into a Z-disk, with the opposite end extending toward the center of the sarcomere, lying in the space between the thick filaments. **Nebulin**, an anchoring protein for actin, coextends with actin and appears to play a regulatory role in mediating actin and myosin interactions ([figure 1.5](#)). Each thin filament contains active sites to which myosin heads can bind.

Actin forms the backbone of the filament. Individual actin molecules are globular proteins (G-actin) and join together to form strands of actin molecules. Two strands then twist into a helical pattern, much like two strands of pearls twisted together.

Tropomyosin is a tube-shaped protein that twists around the actin strands. Troponin is a more complex protein that is attached at regular intervals to both the actin strands and the tropomyosin. This arrangement is depicted in [figure 1.5](#). Tropomyosin and troponin work together in an intricate manner along with calcium ions to maintain relaxation or initiate contraction of the myofibril, which we discuss later in this chapter.

Titin: The Third Myofilament

Titin was not discovered until the late 1970s, well after the sliding filament theory of muscle contraction was proposed. The sliding filament theory adequately describes most functions of muscle during shortening (concentric) and constant-length (isometric) contractions. However, traditional cross-bridge theory does not explain why muscles behave as if they have an internal spring—that is, they produce greater force when stretched (eccentric contractions), a mechanism sometimes called passive force enhancement.⁹ Recent research has determined that titin's stiffness increases with muscle

activation and force development, acting like a spring in active muscle.^{9,14,18}

Titin extends from the Z-disk to the M-band in the sarcomere (figure 1.5). It is attached to the myosin filament in the A-band region, but extends freely in the I-band region, where it functions as a spring. Titin has been known for decades to have structural functions, like keeping myosin aligned during contraction and stabilizing adjacent sarcomeres (figure 1.6). However, it is now known that when skeletal muscles are activated by the release of calcium ions (Ca^{2+}), some calcium binds to titin, changing its stiffness. This helps explain why a muscle's ability to generate more force when stretched is not accounted for by traditional actin–myosin cross-bridge theory.

Further, more recently, when titin is included in three-dimensional models of the sarcomere as a third filament, it becomes clear that filaments do not simply slide but actually twist with each cross-bridge interaction. This has led to a new theory called the winding filament theory that better explains how titin contributes to the force produced by muscle sarcomeres at different lengths.¹⁵ In this updated theory, titin is activated by the calcium ion influx and then winds around the thin filaments, rotating them in the process.



The role of titin in regulating skeletal muscle contractile force helps explain the large increase in force that is observed when muscles are actively stretched. That is, titin is increasingly recognized as a third myofilament that is actively involved in the regulation of skeletal muscle force generation. Among its roles are (1) stabilizing sarcomeres and centering myosin filaments in the middle of the sarcomere, (2) providing increased force when muscles are stretched, and (3) preventing overstretching and damage to the sarcomere by resisting active stretching.⁹

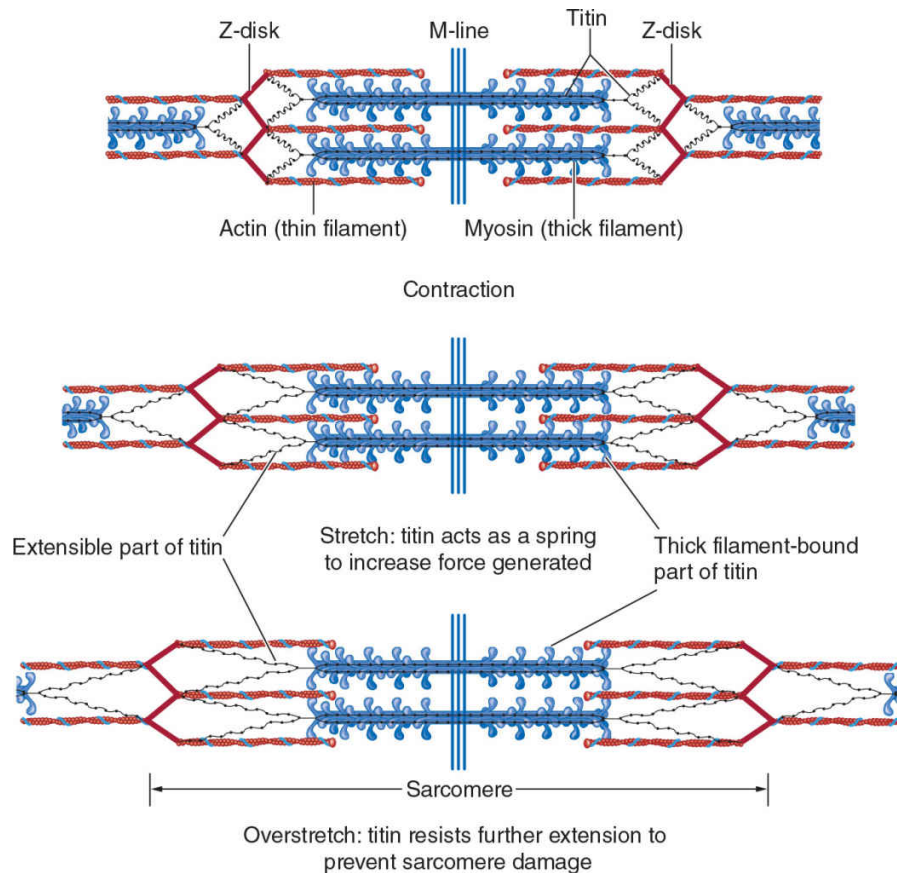


FIGURE 1.6 The mechanism through which the molecule titin acts during muscle contraction. Titin acts as a spring element to increase the force generated and resists overstretch to prevent sarcomere damage.

In Review

- Myofibrils are composed of sarcomeres, the basic contractile units of a muscle.
- A sarcomere is composed of two different-sized filaments, thick and thin filaments, which are responsible for muscle contraction.
- Myosin, the primary protein of the thick filament, is composed of two protein strands, each folded into a globular head at one end.
- The thin filament is composed of actin, tropomyosin, and troponin. One end of each thin filament is attached to a Z-disk.
- A third microfilament, titin, helps stabilize sarcomeres, provides increased force when muscles are stretched, and prevents overstretching and damage to the sarcomere.

Muscle Fiber Contraction

The initiation of contraction of a skeletal muscle occurs in response to a signal from the nervous system. An **α -motor neuron** is a nerve cell that connects with and innervates many muscle fibers. A single α -motor neuron and all the muscle fibers it directly signals are collectively termed a **motor unit** (see [figure 1.7](#)). The synapse or gap between the α -motor neuron and a muscle fiber is referred to as a neuromuscular junction. This is where communication between the nervous and muscular systems occurs.

Excitation–Contraction Coupling

The complex sequence of events that triggers a muscle fiber to contract is termed **excitation–contraction coupling** because it begins with the excitation of a motor nerve and results in contraction of the muscle fibers. The process, depicted in [figure 1.8](#), is initiated by a nerve impulse, or **action potential**, from the brain or spinal cord to an α -motor neuron. The action potential arrives at the α -motor neuron's dendrites, specialized receptors on the neuron's cell body. From here, the action potential travels down the axon to the axon terminals, which are located very close to the plasmalemma.

RESEARCH PERSPECTIVE 1.2

Curving Muscle Fascicles

Muscle fascicles are often drawn in a straight line for ease of illustration. Past experimental measures of muscle fascicle characteristics were based on the idea that the muscle fascicles were straight. However, within the muscle, they are actually curved structures, and the curvature of the fascicles is now recognized as an important characteristic relative to muscle function. Two-dimensional (2D) modeling studies demonstrate that muscle fascicles take on a curved path to provide mechanical stability within the muscle, particularly during contraction. Muscle fascicles curve around regions of the muscle that generate high pressures; thus, they curve more where the largest contractions occur. These 2D models also hint that the curvature could extend into three dimensions (3D), but, until recently, this possibility had not been examined during active muscle contraction.

Using sophisticated imaging techniques, researchers have recently quantified 3D fascicle curvature in triceps surae muscles during contractions at different muscle lengths and torques.¹⁶ Fascicle curvatures increased as the muscle contracted more, indicating an increase in intramuscular pressure at greater levels of contraction. Because this study utilized new 3D imaging approaches, the researchers were able to identify details about the fascicle

curvature that were not detectable in 2D. This more detailed and precise interpretation of the noted 3D fascicle curvature parameters aids our understanding of how pressure is developed in the contracting muscle and of overall muscle function.

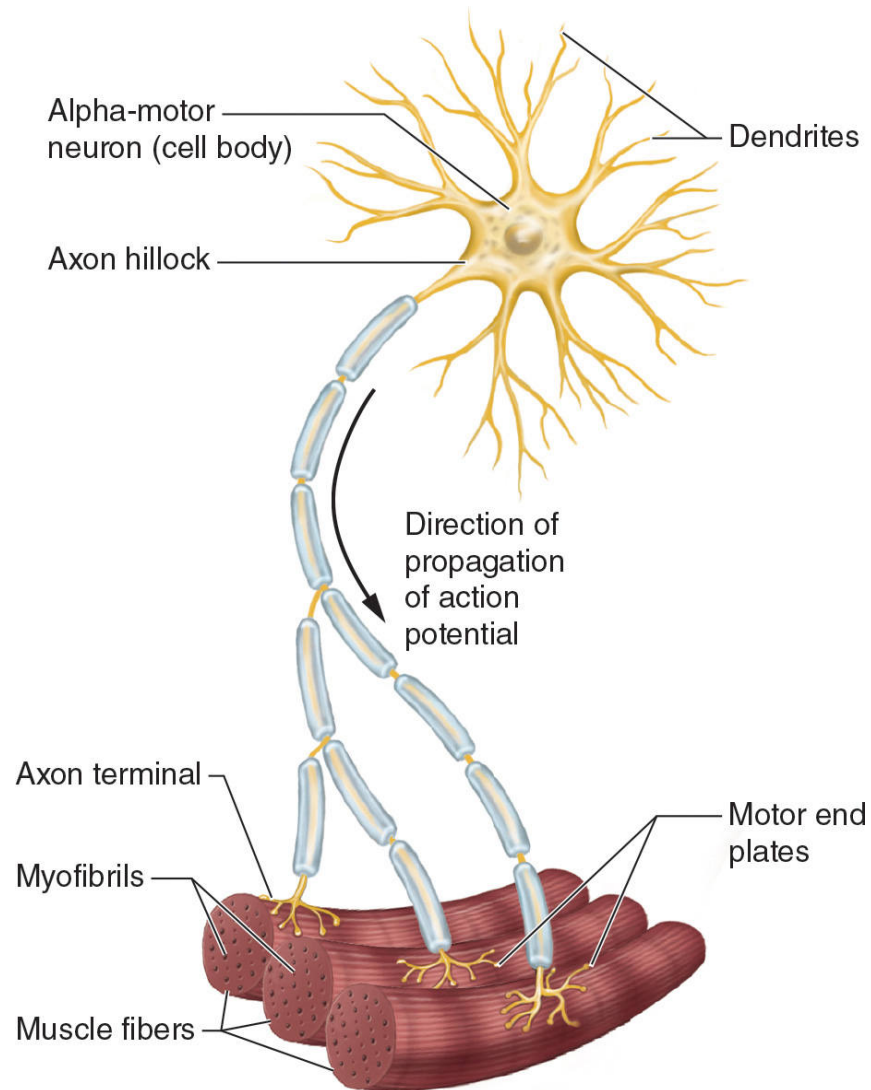


FIGURE 1.7 A motor unit includes one α -motor neuron and all of the muscle fibers it innervates.



When the action potential arrives at the axon terminals, these nerve endings release a signaling molecule or neurotransmitter called acetylcholine (ACh), which crosses the synaptic cleft and binds to receptors on the plasmalemma (see [figure 1.8a](#)). If enough ACh binds to the receptors, the action potential will be transmitted the full length of the muscle fiber as ion gates open in the muscle cell membrane and allow sodium to enter. This process is referred to as depolarization. An action potential must be generated in the muscle cell before the muscle cell can act. These neural events are discussed more fully in [chapter 3](#).

Role of Calcium in the Muscle Fiber

In addition to depolarizing the fiber membrane, the action potential travels over the fiber's network of tubules (T-tubules) to the interior of the cell. The arrival of an electrical charge causes the adjacent SR to release large quantities of stored calcium ions (Ca^{2+}) into the sarcoplasm (see [figure 1.8b](#)).

In the resting state, tropomyosin molecules cover the myosin-binding sites on the actin molecules, preventing the binding of the myosin heads. Once calcium ions are released from the SR, they bind to the troponin on the actin molecules. Troponin, with its strong affinity for calcium ions, is believed to then initiate the contraction process by moving the tropomyosin molecules off the myosin-binding sites on the actin molecules. This is shown in [figure 1.8c](#). Because tropomyosin normally covers the myosin-binding sites, it blocks the attraction between the **myosin cross-bridges** and actin molecules. However, once the tropomyosin has been lifted off the binding sites by troponin and calcium, the myosin heads can attach to the binding sites on the actin molecules.

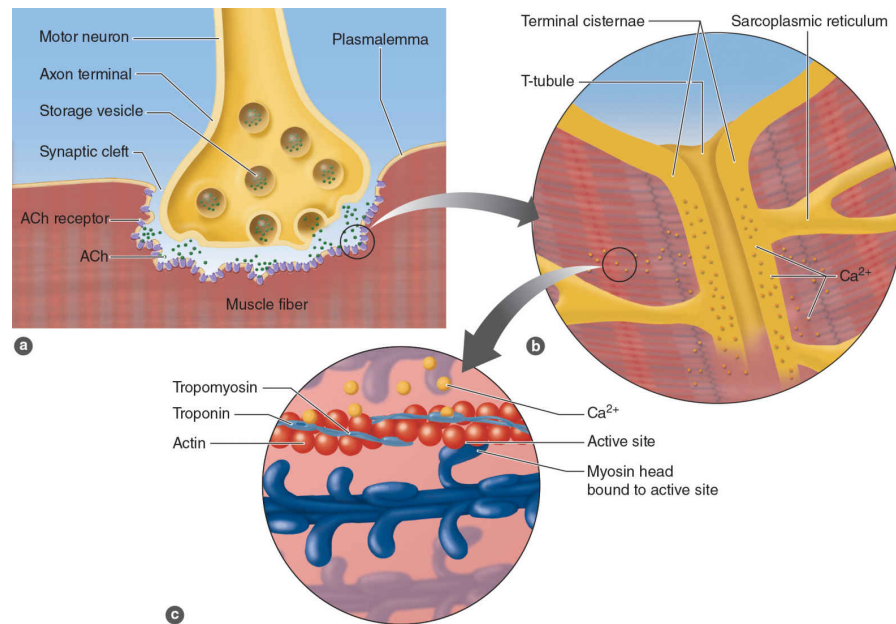
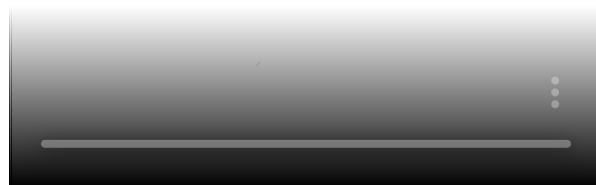


FIGURE 1.8 The sequence of events leading to muscle action, known as excitation–contraction coupling. (a) In response to an action potential, a motor neuron releases acetylcholine (ACh), which crosses the synaptic cleft and binds to receptors on the plasmalemma. If enough ACh binds, an action potential is generated in the muscle fiber. (b) The action potential triggers the release of calcium ions (Ca^{2+}) from the terminal cisternae of the sarcoplasmic reticulum into the sarcoplasm. (c) The Ca^{2+} binds to troponin on the actin filament, and the troponin pulls tropomyosin off the active sites, allowing myosin heads to attach to the actin filament.



The Sliding Filament Theory: How Muscles Create Movement

When muscle contracts, muscle fibers shorten. How do they shorten? The explanation for this phenomenon is termed the **sliding filament theory**. When the myosin cross-bridges are activated, they bind with actin, resulting in a conformational change in the cross-bridge, which causes the myosin head to tilt and to drag the thin filament toward the center of the sarcomere (see [figures 1.9](#) and [1.10](#)). This tilting of the head is referred to as the **power stroke**. The pulling of the thin

filament past the thick filament shortens the sarcomere and generates force. When the fibers are not contracting, the myosin head remains in contact with the actin molecule, but the molecular bonding at the site is weakened or blocked by tropomyosin.

Immediately after the myosin head tilts, it breaks away from the active site, rotates back to its original position, and attaches to a new active site farther along the actin filament. Repeated attachments and power strokes cause the filaments to slide past one another, giving rise to the term *sliding filament theory*. This process continues until the ends of the myosin filaments reach the Z-disks, or until the Ca^{2+} is pumped back into the SR. During this sliding (contraction), the thin filaments move toward the center of the sarcomere and protrude into the H-zone, ultimately overlapping. When this occurs, the H-zone is no longer visible.

Recall that the sarcomeres are joined end to end within a myofibril. Because of this anatomical arrangement, as sarcomeres shorten, the myofibril shortens, causing muscle fibers within a fascicle to shorten. The end result of many such fibers shortening is an organized muscle contraction.

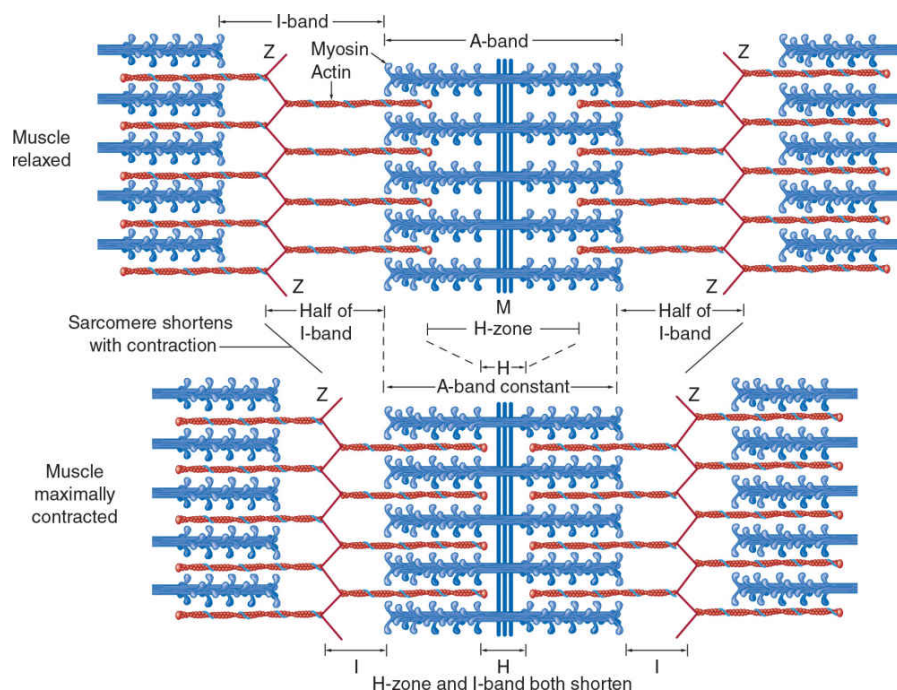


FIGURE 1.9 A sarcomere in its relaxed (top) and contracted (bottom) state, illustrating the sliding of the actin and myosin filaments with contraction.



In Review

- The sequence of events that starts with a motor nerve impulse and results in muscle contraction is called excitation–contraction coupling.
- Muscle contraction is initiated by an α -motor neuron impulse or action potential. The motor neuron releases ACh, which opens up ion gates in the muscle cell membrane, allowing sodium to enter the muscle cell (depolarization). If the cell is sufficiently depolarized, an action potential is generated and muscle contraction occurs.
- When an α -motor neuron is activated, all of the muscle fibers in its motor unit are stimulated to contract.
- The action potential travels along the plasmalemma, then moves through the T-tubule system, causing stored calcium ions to be released from the SR.
- Calcium ions bind with troponin. Then troponin moves the tropomyosin molecules off of the myosin-binding sites on the actin molecules, opening these sites to allow the myosin heads to bind to them.
- Once a strong binding state is established with actin, the myosin head tilts, pulling the thin filament past the thick filament. The tilting of the myosin head is the power stroke.
- Energy is required for muscle contraction to occur. The myosin head binds to the high-energy molecule ATP, and ATPase on the head splits ATP into ADP and P_i , releasing energy to fuel the contraction.
- The end of muscle contraction is signaled when neural activity ceases at the neuromuscular junction. Calcium is actively pumped out of the sarcoplasm and back into the SR for storage. Tropomyosin moves to cover active sites on actin molecules, leading to relaxation between the myosin heads and the binding sites.
- Like muscle contraction, muscle relaxation requires energy supplied by ATP.

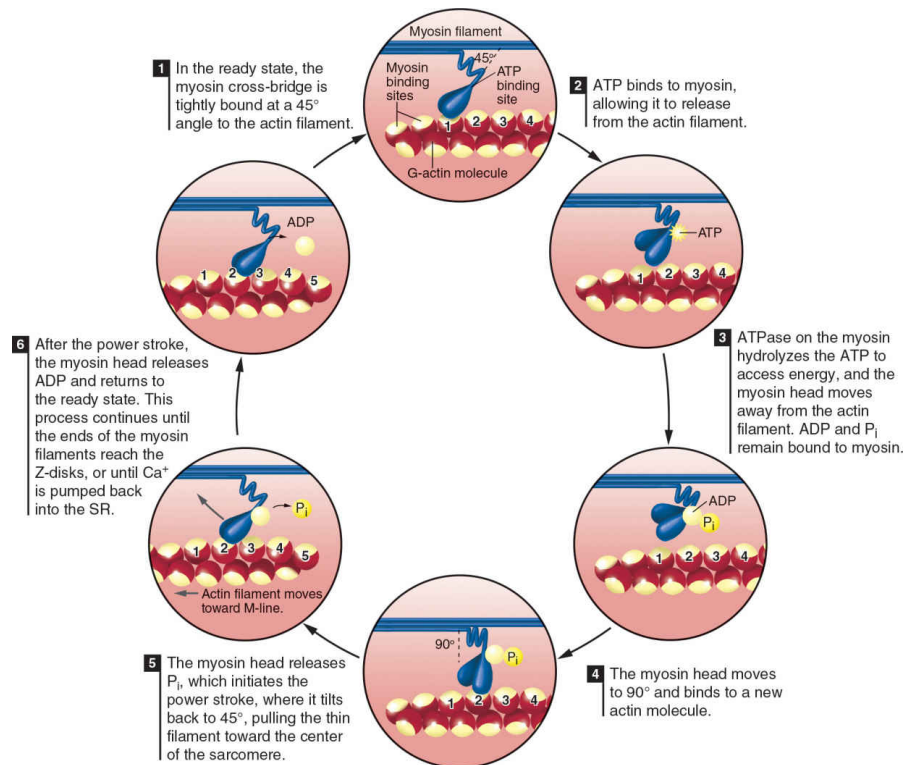


FIGURE 1.10 The molecular events of a contractile cycle illustrating the changes in the myosin head during various phases of the power stroke.



Energy for Muscle Contraction

Muscle contraction is an active process, meaning that it requires energy. In addition to the binding site for actin, a myosin head contains a binding site for the molecule **adenosine triphosphate (ATP)**. The myosin molecule must bind with ATP for muscle contraction to occur because ATP supplies the needed energy.

The enzyme **adenosine triphosphatase (ATPase)**, which is located on the myosin head, splits the ATP to yield adenosine

diphosphate (ADP), inorganic phosphate (P_i), and energy. The energy released from this breakdown of ATP is used to power the tilting of the myosin head. Thus, ATP is the chemical source of energy for muscle contraction. This process is discussed in much more detail in [chapter 2](#).

Muscle Relaxation

Muscle contraction continues as long as calcium is available in the sarcoplasm. At the end of a muscle contraction, calcium is pumped back into the SR, where it is stored until a new action potential arrives at the muscle fiber membrane. Calcium is returned to the SR by an active calcium-pumping system. This is another energy-demanding process that also relies on ATP. Thus, energy is required for both the contraction and relaxation phases.

When the calcium is pumped back into the SR, troponin and tropomyosin return to the resting conformation. This blocks the linking of the myosin cross-bridges and actin molecules and stops the use of ATP. As a result, the thick and thin filaments return to their original relaxed state.

Muscle Fiber Types

Not all muscle fibers are alike. A single skeletal muscle contains fibers having different speeds of shortening and ability to generate maximal force: type I (also called slow or slow-twitch) fibers and type II (also called fast or fast-twitch) fibers. **Type I fibers** take approximately 110 ms to reach peak tension when stimulated. **Type II fibers**, on the other hand, can reach peak tension in about 50 ms. While the terms *slow twitch* and *fast twitch* continue to be used, scientists now prefer to use the terminology *type I* and *type II*, as is the case in this textbook.

Although only one form of type I fiber has been identified, type II fibers can be further classified. In humans, the two major forms of type II fibers are fast-twitch type a (type IIa) and fast-twitch type x (type IIx). [Figure 1.11](#) is a micrograph of human muscle in which thinly sliced (10 μm) cross sections of a muscle sample have been chemically stained to differentiate the fiber types. The type I fibers are stained black; type IIa fibers are unstained and appear white; and

type IIx fibers appear gray. Although not apparent in this figure, a third subtype of fast-twitch fibers has also been identified: type IIc.

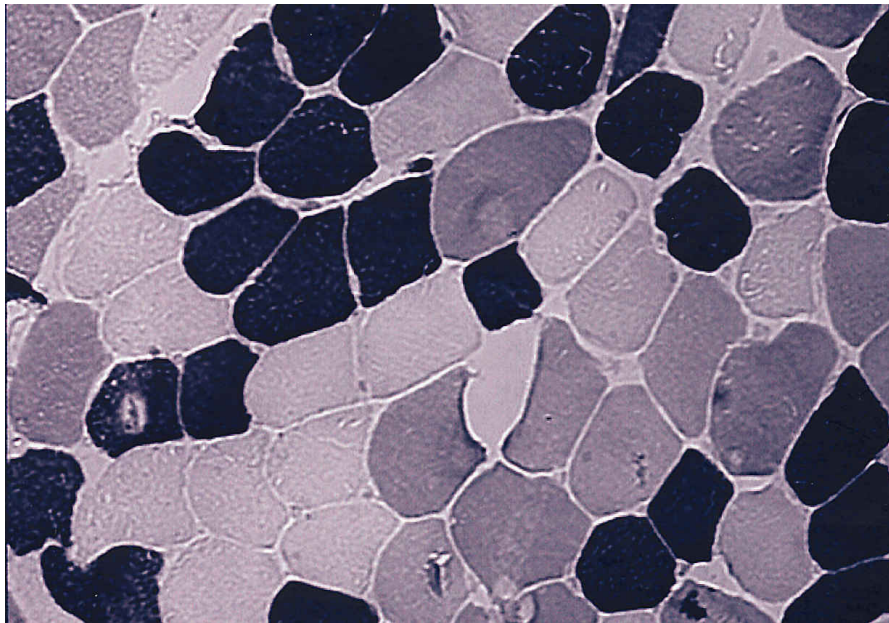


FIGURE 1.11 A photomicrograph showing type I (black), type IIa (white), and type IIx (gray) muscle fibers.



The differences in the type IIa, type IIx, and type IIc fibers are not fully understood, but type IIa fibers are believed to be the most frequently recruited. Only type I fibers are recruited more frequently than type IIa fibers. Type IIc fibers are the least often used. On average, most muscles are composed of roughly 50% type I fibers and 25% type IIa fibers. The remaining 25% are mostly type IIx, with type IIc fibers making up only 1% to 3% of the muscle. Because knowledge about type IIc fibers is limited, we will not discuss them further. The exact percentage of each of these fiber types varies greatly in various muscles and among individuals, so the numbers listed here are only averages. This extreme variation is most evident in athletes, as we will see later in this chapter when we compare fiber types in athletes across sports and events within sports.

In the early 1900s, a needle biopsy procedure was developed to study muscular dystrophy. In the 1960s, this technique was adapted to sample muscle for studies in exercise physiology, specifically to help determine muscle fiber types.

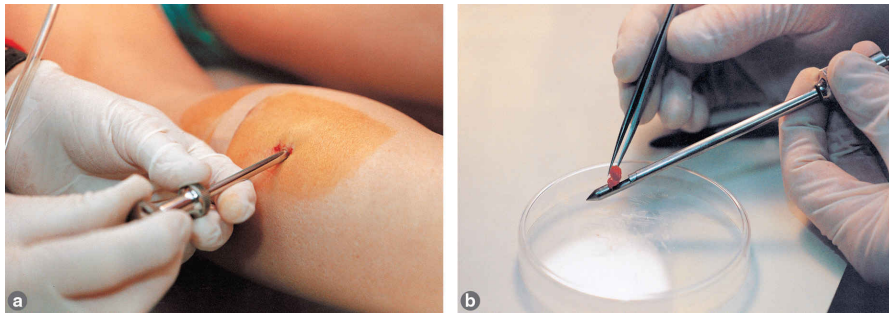


FIGURE 1.12 (a) The use of a biopsy needle to obtain a sample from the leg muscle of an elite female runner. (b) A close-up view of a muscle biopsy needle and a small piece of muscle tissue.

A muscle biopsy ([figure 1.12](#)) involves removing a very small piece of muscle tissue from the muscle belly for analysis. The area from which the sample is taken is first numbed with a local anesthetic, and then a small incision (approximately 1 cm, or 0.4 in.) is made with a scalpel through the skin, subcutaneous tissue, and connective tissue. A hollow needle is then inserted to the appropriate depth into the belly of the muscle. A small plunger is pushed through the center of the needle to snip off a very small sample of muscle. The biopsy needle is withdrawn, and the sample, weighing 10 to 100 mg, is removed, cleaned of blood, mounted, and quickly frozen. It is then thinly sliced, stained, and examined under a microscope.

This method allows us to study muscle fibers and gauge the effects of acute exercise and chronic training on fiber composition. Microscopic and biochemical analyses of the samples aid our understanding of the muscles' ability to produce energy for contraction.

Characteristics of Type I and Type II Fibers

Different muscle fiber types play different roles in exercise and sport. This is largely due to differences in their inherent characteristics.

ATPase

Type I and type II fibers differ in their speed of contraction. This difference results primarily from different forms of myosin ATPase. Recall that myosin ATPase is the enzyme that splits ATP to release energy to drive contraction. Type I fibers have a slow form of myosin ATPase, whereas type II fibers have a fast form. In response to neural stimulation, ATP is split more rapidly in type II fibers than in type I fibers. As a result, cross-bridges cycle more rapidly in type II fibers.

One of the methods used to classify muscle fibers is a chemical staining procedure applied to a thin slice of tissue. This staining technique measures the ATPase activity in the fibers. Thus, the type I, type IIa, and type IIx fibers stain differently, as depicted in [figure 1.11](#). This technique makes it appear that each muscle fiber has only one type of ATPase, but fibers can have a mixture of ATPase types. Some have a predominance of type I ATPase, but others have mostly type II ATPase. Their appearance in a stained slide preparation should be viewed as a continuum rather than as absolutely distinct types.

A newer method for identifying fiber types is to chemically separate the different types of myosin molecules (isoforms) by using a process called gel electrophoresis. In electrophoresis, the isoforms are separated by weight in an electric field to show the bands of protein (i.e., myosin) that characterize type I, type IIa, and type IIx fibers. Although our discussion here categorizes fiber types simply as slow twitch (type I) and fast twitch (type IIa and type IIx), scientists have further subdivided these fiber types. The use of electrophoresis has led to the detection of myosin hybrids or fibers that possess two or more forms of myosin. With this method of analysis, the fibers are classified as I, Ic (I/IIa), IIc (IIa/I), IIa, IIax, IIxa, and IIx.¹² In this book, we will use the histochemical method of identifying fibers by their primary isoforms, types I, IIa, and IIx.

[Table 1.1](#) summarizes the characteristics of the different muscle fiber types. The table also includes alternative names that are used in other classification systems to refer to the various muscle fiber types.

TABLE 1.1 Classification of Muscle Fiber Types	
Fiber classification	

System 1 (preferred)	Type I	Type IIa	Type IIx
System 2	Slow twitch (ST)	Fast-twitch a (FTa)	Fast-twitch x (FTx)
System 3	Slow oxidative (SO)	Fast oxidative/glycolytic (FOG)	Fast glycolytic (FG)
Characteristics of fiber types			
Oxidative capacity	High	Moderately high	Low
Glycolytic capacity	Low	High	Highest
Contractile speed	Slow	Fast	Fast
Fatigue resistance	High	Moderate	Low
Motor unit strength	Low	High	High

Sarcoplasmic Reticulum

Type II fibers have a more highly developed SR than do type I fibers. Thus, type II fibers are more adept at delivering calcium into the muscle cell when stimulated. This ability is thought to contribute to the faster speed of contraction (V_o) of type II fibers. On average, human type II fibers have a V_o that is five to six times faster than that of type I fibers. Although the amount of force (P_o) generated by type II and type I fibers with the same diameter is about the same, the calculated power ($\mu\text{N} \cdot \text{fiber length}^{-1} \cdot \text{s}^{-1}$) of a type II fiber is three to five times greater than that of a type I fiber because of a faster shortening velocity. This may explain in part why individuals who have a predominance of type II fibers in their leg muscles tend to be better sprinters than individuals who have a high percentage of type I fibers, all other things being equal.

RESEARCH PERSPECTIVE 1.3

More About Titin

Eccentric contractions are those during which active muscles are stretched or elongated, such as during the lowering of a weight by the biceps or walking down stairs. Unlike for concentric contractions, the classic theories of muscle contraction—the sliding filament and cross-bridge theories—do not agree well with some aspects of eccentrically contracting muscles. In the cross-bridge model, greater force is developed during an eccentric contraction than for a corresponding isometric or concentric contraction because the attached cross-bridges are more strained. However, skeletal muscles are known to also have history-dependent properties; that is, muscles behave differently depending on preceding contractions. For example, when isometric contractions follow an eccentric contraction, they often demonstrate a longer-lasting, steady-state isometric force compared with the force produced during an isometric contraction not preceded by an eccentric contraction. The classic cross-bridge model involving actin and myosin does not explain such history-dependent

properties of skeletal muscle, necessitating a careful reassessment of the currently accepted theory of muscle contraction.

A group of researchers recently proposed a new mechanism that better explains the history-dependent properties of muscle and eccentric contractions by adding a small component to the classic cross-bridge theory.¹⁰ This tweak includes a new key role for the structural protein titin, whose stiffness and force are adjusted upon activation and force production. In this new model, muscle can be stretched passively against little resistance, but upon activation, titin-based force becomes dominant and contributes to eccentric force.

Titin's traditional role (discussed in this chapter) has been associated with centering the thick filaments in the sarcomere and preventing sarcomeres from becoming overstretched by putting the brakes on active stretching. Yet another function of titin is to serve as the third sarcomere myofilament, contributing to active force production during and following eccentric contractions. In this role, titin appears to be critical in the residual force enhancement in skeletal muscle after an eccentric contraction. This theory is based on research evidence that experimentally eliminating titin in single myofibrils abolishes all force transmission across sarcomeres and all residual and passive force enhancement. Thus, titin acts as a kind of molecular spring that increases the muscle's stiffness, and thus its force, in active compared with passive muscle contraction. Researchers now speculate that titin contributes to active force production by changing its stiffness (i.e., when titin becomes stiffer, it can produce more force). The researchers theorize that titin's stiffness increases (1) by binding to calcium upon activation and (2) by binding to actin and decreasing its length.

Although preliminary results and theoretical models provide support for this new three-filament model of force production, the molecular details have not been worked out. If correct, it would provide a substantial update to the classic cross-bridge theory. If proven, the new three-filament model of muscle contraction would simultaneously add to our understanding of eccentric contractions and explain the history-dependent properties of muscle, information not previously explainable using the classical two-filament cross-bridge theory.

Motor Units

Recall that a motor unit is composed of a single α -motor neuron and the muscle fibers it innervates. The α -motor neuron appears to determine whether the fibers are type I or type II. The α -motor neuron in a type I motor unit has a smaller cell body and typically innervates a cluster of ≤ 300 muscle fibers. In contrast, the α -motor neuron in a type II motor unit has a larger cell body and innervates ≥ 300 muscle fibers. This difference in the size of motor units means that when a

single type I α -motor neuron stimulates its fibers, far fewer muscle fibers contract than when a single type II α -motor neuron stimulates its fibers. Consequently, type II muscle fibers reach peak tension faster and collectively generate more force than type I fibers. The difference in maximal isometric force development between type II and type I motor units is attributable to two characteristics: the number of muscle fibers per individual motor unit and the difference in the size of type II and type I fibers. Type I and type II fibers of the same diameter generate about the same force. On average, however, type II fibers tend to be larger than type I fibers, and type II motor units tend to have more muscle fibers than do the type I motor units.

Distribution of Fiber Types

As mentioned earlier, the percentages of type I and type II fibers are not the same in all the muscles of the body. Generally, arm and leg muscles have similar fiber compositions within an individual. An endurance athlete with a predominance of type I fibers in his or her leg muscles will likely have a high percentage of type I fibers in the arm muscles as well. A similar relationship exists for type II fibers. There are some exceptions, however. The soleus muscle (beneath the gastrocnemius in the calf), for example, is composed of a very high percentage of type I fibers in everyone.

Fiber Type and Exercise

Because of these differences in type I and type II fibers, one might expect that these fiber types would also have different functions when people are physically active. Indeed, this is the case.

Type I Fibers

In general, type I muscle fibers have a high level of aerobic endurance. Aerobic means “in the presence of oxygen,” so oxidation is an aerobic process. Type I fibers are very efficient at producing ATP from the oxidation of carbohydrate and fat, which is discussed in [chapter 2](#).

Recall that ATP is required to provide the energy needed for muscle fiber contraction and relaxation. As long as oxidation occurs, type I fibers continue producing ATP, allowing the fibers to remain active. The ability to maintain muscular activity for a prolonged period

is known as muscular endurance, so type I fibers have high aerobic endurance. Because of this, they are recruited most often during low-intensity endurance events (e.g., marathon running) and during most daily activities for which the muscle force requirements are low (e.g., walking).

Type II Fibers

Type II muscle fibers, on the other hand, have relatively poor aerobic endurance when compared to type I fibers. They are better suited to perform anaerobically (without oxygen). This means that in the absence of adequate oxygen, ATP is formed through anaerobic pathways, not oxidative pathways. (We discuss these pathways in detail in [chapter 2](#).)

Type IIa motor units generate considerably more force than do type I motor units, but type IIa motor units also fatigue more easily because of their limited endurance. Thus, type IIa fibers appear to be the primary fiber type used during shorter, higher-intensity endurance events, such as the mile run or the 400 m swim.

Although the significance of type IIx fibers is not fully understood, they apparently are not easily activated by the nervous system. Thus, they are used rather infrequently in normal, low-intensity activity but are predominantly used in highly explosive events such as the 100 m dash and the 50 m sprint swim. Characteristics of the various fiber types are summarized in [table 1.2](#).

One of the most advanced methods for the study of human muscle fibers is to dissect fibers out of a muscle biopsy sample, suspend a single fiber between force transducers, and measure its strength and **single-fiber contractile velocity (V_o)**. From [figure 1.13](#), one can see that all of the single fibers tend to reach their peak power when the fibers are generating only about 20% of their peak force. However, it is quite clear that the peak power of the type II fibers is considerably higher than that of the type I fibers.

TABLE 1.2 Structural and Functional Characteristics of Muscle Fiber Types

Characteristic	Fiber type		
	Type I	Type IIa	Type IIx
Fibers per motor neuron	≤300		

Motor neuron size	Smaller	≥300	≥300
Motor neuron conduction velocity	Slower	Larger	Larger
Contraction speed (ms)	110	Faster	Faster
Type of myosin ATPase	Slow	50	50
Sarcoplasmic reticulum development	Low	Fast	Fast
		High	High

Determination of Fiber Type

The characteristics of muscle fibers appear to be determined early in life, perhaps within the first few years. Studies with identical twins have shown that muscle fiber type, for the most part, is genetically determined, changing little from childhood to middle age. These studies reveal that identical twins have nearly identical proportions of fiber types, whereas fraternal twins differ in their fiber type profiles. The genes we inherit from our parents likely determine which α -motor neurons innervate our individual muscle fibers. After innervation is established, muscle fibers differentiate (become specialized) according to the type of α -motor neuron that stimulates them. Some recent evidence, however, suggests that endurance training, strength training, and muscular inactivity may cause a shift in the myosin isoforms. Consequently, training may induce a small change, perhaps less than 10%, in the percentage of type I and type II fibers. Further, both endurance and resistance training have been shown to reduce the percentage of type IIx fibers while increasing the fraction of type IIa fibers.

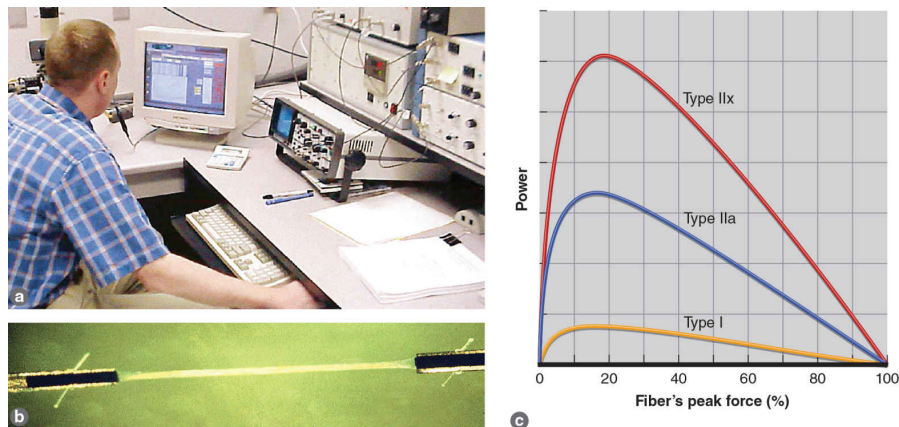


FIGURE 1.13 (a) The dissection and (b) suspension of a single muscle fiber to study the physiology of different fiber types. (c) Differences in peak power generated by each fiber type at various percentages of maximal force.

Studies of older men and women have shown that aging may alter the distribution of type I and type II fibers. As we grow older, muscles tend to lose type II motor units, which increases the percentage of type I fibers.

In Review

- Most skeletal muscles contain both type I and type II fibers.
- Different fiber types have different myosin ATPase activities. The ATPase in the type II fibers acts faster than the ATPase in type I fibers.
- Type II fibers have a more highly developed SR, enhancing the delivery of calcium needed for muscle contraction.
- α -Motor neurons innervating type II motor units are larger and innervate more fibers than do α -motor neurons for type I motor units. Thus, type II motor units have more (and larger) fibers to contract and can produce more force than type I motor units.
- The proportions of type I and type II fibers in a person's arm and leg muscles are usually similar.
- Type I fibers have higher aerobic endurance and are well suited to low-intensity endurance activities.
- Type II fibers are better suited for anaerobic activity. Type IIa fibers play a major role in high-intensity exercise. Type IIx fibers are activated when the force demanded of the muscle is high.

Skeletal Muscle and Exercise

Having reviewed the overall structure of muscle, the process by which it develops force, and the types of muscle fibers, we now look more specifically at how muscle functions during exercise. Strength, endurance, and speed depend largely on the muscle's ability to produce energy and force. This section examines how muscle accomplishes this task.

Muscle Fiber Recruitment

When an α -motor neuron carries an action potential to the muscle fibers in the motor unit, all fibers in the unit develop force. Activating more motor units is the way muscles produce more force. When little force is needed, only a few motor units are recruited. Recall from our earlier discussion that type IIa and type IIx motor units contain more

muscle fibers than type I motor units do. Skeletal muscle contraction involves a progressive recruitment of type I, followed by type II motor units, depending on the requirements of the activity being performed. As the intensity of the activity increases, the number of fibers recruited increases in the following order, in an additive manner: type I → type IIa → type IIx.

Motor units are generally activated on the basis of a fixed order of fiber recruitment. This is known as the **principle of orderly recruitment**, in which the motor units within a given muscle appear to be ranked. Let's use the biceps brachii as an example: Assume a total of 200 motor units, which are ranked on a scale from 1 to 200. For an extremely fine muscle contraction requiring very little force production, the motor unit ranked number 1 would be recruited. As the requirements for force production increase, numbers 2, 3, 4, and so on would be recruited, up to a maximal muscle contraction that would activate most, if not all, of the motor units. For the production of a given force, the same motor units are usually recruited each time and in the same order.

A mechanism that may partially explain the principle of orderly recruitment is the **size principle**, which states that the order of recruitment of motor units is directly related to the size of their motor neuron. Motor units with smaller motor neurons will be recruited first. Because the type I motor units have smaller motor neurons, they are the first units recruited in graded movement (going from very low to very high rates of force production). The type II motor units then are recruited as the force needed to perform the movement increases. It is unclear at this time how the size principle relates to complex athletic movements.

During events that last several hours, exercise is performed at a submaximal pace, and the tension in the muscles is relatively low. As a result, the nervous system tends to recruit those muscle fibers best adapted to endurance activity: the type I and some type IIa fibers. As the exercise continues, these fibers become depleted of their primary fuel supply (glycogen), and the nervous system must recruit more type IIa fibers to maintain muscle tension. Finally, when the type I and type IIa fibers become exhausted, the type IIx fibers may be recruited to continue exercising.

In Review

- Motor units give all-or-none responses. Activating more motor units produces more force.
- In low-intensity activity, most muscle force is generated by type I fibers. As the intensity increases, type IIa fibers are recruited, and at even higher intensities, the type IIx fibers are activated. The same pattern of recruitment is followed during events of long duration.

This may explain why fatigue seems to come in stages during events such as the marathon, a 42 km (26.1 mi) run. It also may explain why it takes great conscious effort to maintain a given pace near the finish of the event. This conscious effort results in the activation of muscle fibers that are not easily recruited. Such information is of practical importance to our understanding of the specific requirements of training and performance.

Fiber Type and Athletic Success

From what we have just discussed, it appears that athletes who have a high percentage of type I fibers might have an advantage in prolonged endurance events, whereas those with a predominance of type II fibers could be better suited for high-intensity, short-term, and explosive activities. But does the relative proportion of an athlete's muscle fiber types determine athletic success?

The muscle fiber makeup of successful athletes from a variety of athletic events and of nonathletes is shown in [table 1.3](#). As anticipated, the leg muscles of distance runners, who rely on endurance, have a predominance of type I fibers.⁴ Studies of elite male and female distance runners revealed that many of these athletes' gastrocnemius (calf) muscles may contain more than 90% type I fibers. World champions in the marathon are reported to possess 93% to 99% type I fibers in their gastrocnemius muscles. In contrast, the gastrocnemius muscles are composed principally of type II fibers in sprint runners, who rely on speed and strength. World-class sprinters have only about 25% type I fibers in this muscle. Also, although muscle fiber cross-sectional area varies markedly among elite distance runners, type I fibers in their leg muscles average about 22% more cross-sectional area than type II fibers.^{5,6} Swimmers tend

to have higher percentages of type I fibers (60%-65%) in their arm muscles than untrained subjects (45%-55%).

The fiber composition of muscles in distance runners and sprinters is markedly different. However, it may be a bit risky to think we can select champion distance runners and sprinters solely on the basis of predominant muscle fiber type. Other factors, such as cardiovascular function, motivation, training, and muscle size, also contribute to success in such events of endurance, speed, and strength. Thus, fiber composition alone is not a reliable predictor of athletic success.

Muscle Contraction

We have examined the different muscle fiber types. We understand that all fibers in a motor unit, when stimulated, act at the same time and that different fiber types are recruited in stages, depending on the force required to perform an activity. Now we can turn our attention to how whole muscles work to produce movement.

TABLE 1.3 Percentages and Cross-Sectional Areas of Type I and Type II Fibers in Selected Muscles of Male and Female Athletes

	Sex	Muscle	% type I	% type II	Cross-sectional area (mm ²)	
					Type I	Type II
Sprint runners	M	Gastrocnemius	24	76	5,878	6,034
	F	Gastrocnemius	27	73	3,752	3,930
Distance runners	M	Gastrocnemius	79	21	8,342	6,485
	F	Gastrocnemius	69	31	4,441	4,128
Cyclists	M	Vastus lateralis	57	43	6,333	6,116
	F	Vastus lateralis	51	49	5,487	5,216
Swimmers	M	Posterior deltoid	67	33		
Weightlifters	M	Gastrocnemius	44	56	5,060	8,910
	M	Deltoid	53	47	5,010	8,450
Triathletes	M	Posterior deltoid	60	40		
	M	Vastus lateralis	63	37		
	M	Gastrocnemius	59	41		
Canoeists	M	Posterior deltoid	71	29	4,920	7,040
Shot-putters	M	Gastrocnemius	38	62	6,367	6,441
Nonathletes	M	Vastus lateralis	47	53	4,722	4,709
	F	Gastrocnemius	52	48	3,501	3,141

Types of Muscle Contraction

Muscle movement generally can be categorized into three types of contractions—concentric, static, and eccentric. In many activities,

such as running and jumping, all three types of contraction may occur in the execution of a smooth, coordinated movement. For the sake of clarity, though, we will examine each type separately.

A muscle's principal action, shortening, is referred to as a **concentric contraction**, the most familiar type of contraction. To understand muscle shortening, recall our earlier discussion of how the thin and thick filaments slide across each other. In a concentric contraction, the thin filaments are pulled toward the center of the sarcomere. Because joint movement is produced, concentric contractions are considered **dynamic contractions**.

Muscles can also act without moving. When this happens, the muscle generates force, but its length remains static (unchanged). This is called a **static (isometric) muscle contraction** because the joint angle does not change. A static contraction occurs, for example, when one tries to lift an object that is heavier than the force generated by the muscle, or when one supports the weight of an object by holding it steady with the elbow flexed. In both cases, the person feels the muscles tense, but there is no joint movement. In a static contraction, the myosin cross-bridges form and are recycled, producing force, but the external force is too great for the thin filaments to be moved. They remain in their normal position, so shortening can't occur. If enough motor units can be recruited to produce sufficient force to overcome the resistance, a static contraction can become a dynamic one.

Muscles can exert force even while lengthening. This movement is an **eccentric contraction**. Because joint movement occurs, this is also a dynamic contraction. An example of an eccentric contraction is the action of the biceps brachii when one extends the elbow slowly to lower a heavy weight. In this case, the thin filaments are pulled farther away from the center of the sarcomere, essentially stretching it.

Generation of Force

Whenever muscles contract, whether the contraction is concentric, static, or eccentric, the force developed must be graded to meet the needs of the task or activity. Using golf as an example, the force needed to tap in a 1 m (~39 in.) putt is far less than that needed to drive the ball 250 m (273 yd) from the tee to the middle of the fairway.

The amount of muscle force developed is dependent on the number and type of motor units activated, the frequency of stimulation of each motor unit, the size of the muscle, the muscle fiber and sarcomere length, and the muscle's speed of contraction.

Motor Units and Muscle Size More force can be generated when more motor units are activated. Type II motor units generate more force than type I motor units because a type II motor unit contains more muscle fibers than a type I motor unit. In a similar manner, larger muscles, having more muscle fibers, can produce more force than smaller muscles.

Frequency of Stimulation of the Motor Units: Rate Coding A single motor unit can exert varying levels of force dependent on the frequency at which it is stimulated. This is illustrated in [figure 1.14](#).¹ The smallest contractile response of a muscle fiber or a motor unit to a single electrical stimulus is termed a **twitch**. A series of three stimuli in rapid sequence, before complete relaxation from the first stimulus, can elicit an even greater increase in force or tension. This is termed **summation**. Continued stimulation at higher frequencies can lead to the state of **tetanus**, resulting in the peak force or tension of the muscle fiber or motor unit. **Rate coding** is the term used to refer to the process by which the tension of a given motor unit can vary from that of a twitch to that of tetanus by increasing the frequency of stimulation of that motor unit.

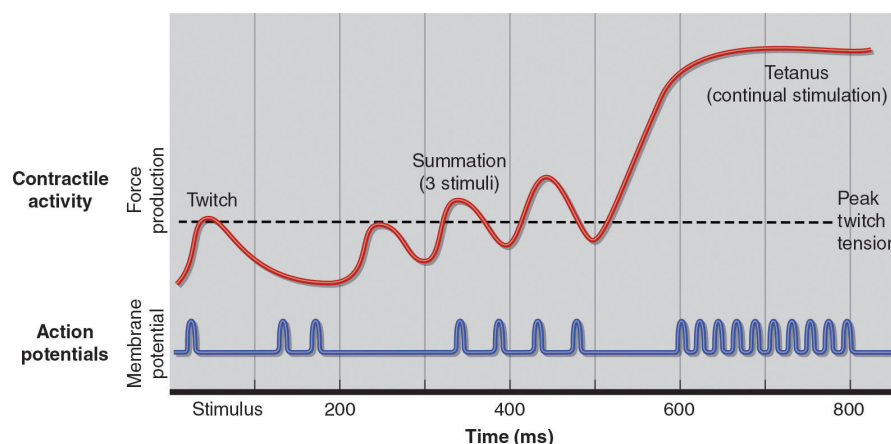


FIGURE 1.14 Variation in force or tension produced based on electrical stimulation frequency, illustrating the concepts of a twitch, summation, and tetanus.

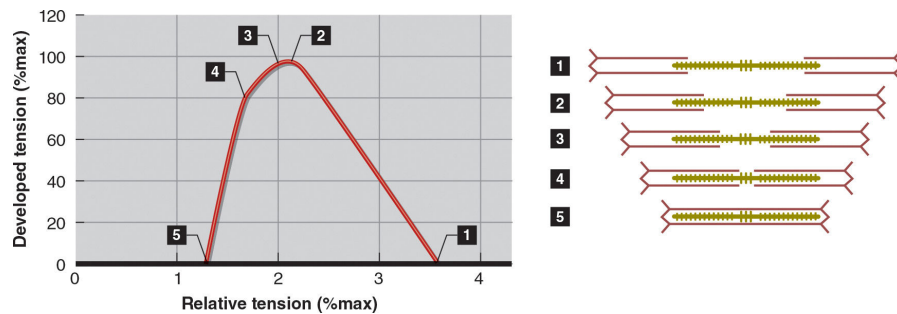


FIGURE 1.15 Variation in force or tension produced (% of maximum) with changes in sarcomere length, illustrating the concept of optimal length for force production.

Adapted by permission from B.R. MacIntosh, P.F. Gardiner, and A.J. McComas, *Skeletal Muscle: Form and Function*, 2nd ed. (Champaign, IL: Human Kinetics, 2006), 156.

Length–Tension Relation Each muscle fiber has an optimal length for generating force. Recall that each muscle fiber is composed of sarcomeres connected end to end and that these sarcomeres are made up of both thick and thin filaments. The optimal sarcomere length is defined as that length where there is optimal overlap between the thick and thin filaments. This maximizes the potential for cross-bridge interaction, as illustrated in [figure 1.15](#).¹² When a sarcomere is overly stretched (1) or shortened (5), little or no force can be developed since there is little cross-bridge interaction. The muscle's resting length is determined by the tendons that attach muscles to the bones on either end. As it turns out, this natural resting length maximizes the muscle's ability to generate force, referred to as the **length–tension relation**.

The implication of this relation is that muscle length, and therefore joint angle, will provide a mechanical advantage for force generation of a particular muscle or muscle group. The length–tension curve

shown in [figure 1.15](#) illustrates this phenomenon. Maximal tension can be achieved at sarcomere lengths between 2.0 and 2.25 μm , where the overlap between myosin and actin is optimal (i.e., the highest number of cross-bridges can be formed). As the sarcomere becomes elongated ($>2.25 \mu\text{m}$), the number of possible cross-bridges decreases, and therefore tension development (the descending limb of the curve) decreases accordingly. When sarcomeres are shortened to lengths $<2.0 \mu\text{m}$, the ability of myosin to interact with actin decreases because there are fewer myosin heads available to interact with actin (actin moves close to the M-line where there are few myosin heads; see [figure 1.5](#)). Another possible explanation for the reduced ability to produce force at lengths $<2.0 \mu\text{m}$ is the physical constraint imposed by myosin reaching the Z-line of the sarcomere.

Force–Velocity Relation The ability of the muscle to develop force also depends on the speed of contraction. When people try to lift a very heavy object, they tend to do it slowly, maximizing the force they can apply to it. If they grab it and quickly try to lift it, they will likely fail, if not injure themselves. The **force–velocity relation** of a muscle illustrates muscle force as a function of the speed of contraction. During concentric (muscle-shortening) contractions, maximal force development decreases progressively as the speed of contraction increases. However, with eccentric (muscle-lengthening) contractions, the opposite is true.

This relation between force development and speed of contraction can be explained by the number of total cross-bridges attached at various speeds of contraction. When a muscle is contracting slowly, there is more time for cross-bridge formation than when contractions occur at higher speeds. In other words, when cross-bridges are formed at higher velocities, the ability of the muscle to produce force is reduced.

The force–velocity relation applies to both shortening and lengthening contractions. As depicted in [figure 1.16](#), increasing the velocity of contraction while shortening (moving rightward along the x-axis) reduces force. Another way to think of the force–velocity relation is in terms of applying an external force to the muscle, such as performing a biceps curl. As the load gets heavier, the speed of contraction gets slower. When the load applied equals the maximal

isometric force of the muscle, contraction velocity equals zero (by definition, an isometric contraction involves no movement). Now, let's explore what will happen when the load applied to the muscle is higher than the maximal isometric force and the muscle lengthens. In this case, the ability of the muscle to produce force will increase as a function of speed (moving leftward along the x-axis in [figure 1.16](#)) because as the load increases beyond maximal isometric, the speed of contraction will also increase.

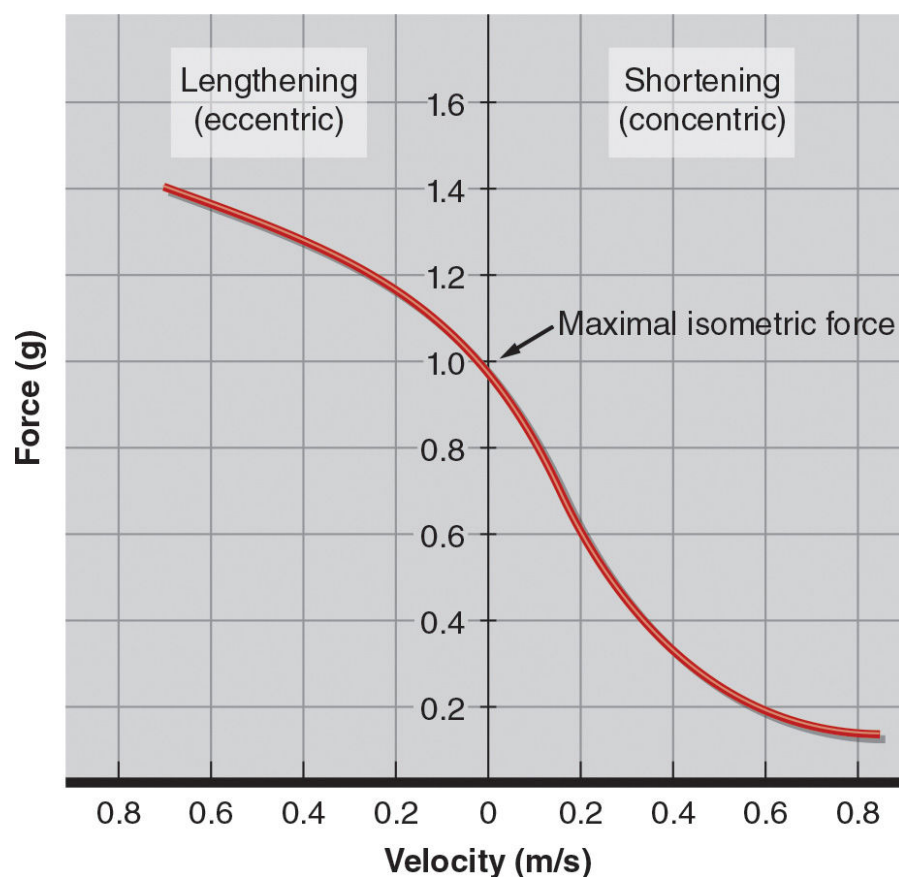


FIGURE 1.16 The relation between muscle lengthening and shortening velocity and force production. Note that the capacity for the muscle to generate force is greater during eccentric (muscle-lengthening) actions than during concentric (muscle-shortening) actions.



Muscle Memory

Muscle force production depends on muscle mass. Because muscle fibers are large, they need evenly distributed multiple nuclei along their length in order to support all of the protein synthesis that occurs within the vast intracellular volume. Muscle fibers constantly change size, getting smaller with disuse (atrophy) and larger with training (hypertrophy). Standard thinking has been that muscle precursor satellite cells, small mononuclear stem cells, multiply during hypertrophy, fuse with existing muscle fibers, and supply additional nuclei as the fibers grow in size. Alternatively, during atrophy, unnecessary nuclei are cleared by a process called apoptosis, or programmed death.

A newer model has emerged that better explains the mechanisms that underlie changes in muscle fiber size and muscle mass.³ In a study by Bruusgaard and colleagues, rat hindlimb muscles were hypertrophied by overloading, and their nuclei were measured by injecting labeled nucleotides. Beginning on day 6, the number of nuclei began to increase, increasing by 54% over 21 days. Fiber cross-sectional area did not begin to increase until day 9. In another group of rats, motor nerves were severed, causing muscle atrophy. The cross-sectional area decreased by 60% of the highest value of the hypertrophied group, but the number of nuclei was unchanged.

In trained individuals, retraining after a period of disuse occurs more quickly than in novice exercisers, and such muscle memory has typically been attributed to neural control of the muscle. It now appears that the nuclei may be the site of such memory. However, as pointed out by Lee and Burd,¹¹ a role for satellite cells in muscle hypertrophy cannot be excluded, and satellite cells undoubtedly contribute to overall skeletal muscle mass. It is possible that satellite cells may be necessary to sustain the mass and may be integral in maintaining muscle quality and function ([figure 1.17](#)).

In Review

- Among elite athletes, muscle fiber type composition differs by sport and event, with speed and strength events characterized by higher percentages of type II fibers and endurance events by higher percentages of type I fibers.
- The three main types of muscle contraction are concentric, in which the muscle shortens; static or isometric, in which the muscle acts but the joint angle is

unchanged; and eccentric, in which the muscle lengthens.

- Force production can be increased both through the recruitment of more motor units and through an increase in the frequency of stimulation (rate coding) of the motor units.
- Force production is maximized at the muscle's optimal length. At this length, the amount of energy stored and the number of linked actin–myosin cross-bridges are optimal.
- Speed of contraction also affects the amount of force produced. For concentric contraction, maximal force is achieved with slower contractions. The closer to zero the velocity (isometric), the more force can be generated. With eccentric contractions, however, faster movement allows more force production.
- In addition to satellite cells, preserving the number of muscle fiber nuclei may help explain why previously trained muscles adapt more quickly to retraining after a period of disuse.

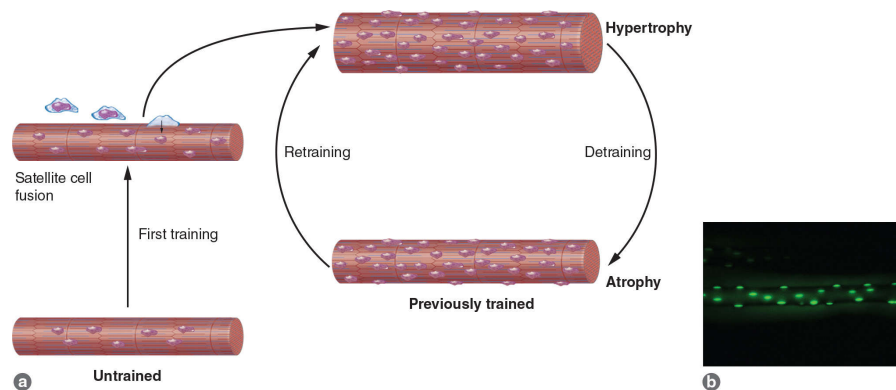


FIGURE 1.17 (a) A model to explain how the nuclei of muscle fibers may be the site of muscle memory. This theory explains why previously trained muscles adapt more quickly to retraining after a period of disuse. (b) Photomicrograph showing peripheral distribution of nuclei within a muscle fiber.

(a) Reprinted from J.C. Bruusgaard et al., "Myonuclei Acquired by Overload Exercise Precede Hypertrophy and are Not Lost on Detraining," *Proceedings of the National Academy of Sciences* 107 (2010): 15111-15116. By permission of J.C. Bruusgaard

IN CLOSING

In this chapter, we reviewed the components of skeletal muscle. We considered the differences in fiber types and their impact on physical performance. We learned how muscles generate force and produce movement. Now that we understand how movement is produced, we turn our attention to how movement is fueled. In the next chapter, we focus on metabolism and energy production.

KEY TERMS

actin
action potential
adenosine triphosphatase (ATPase)
adenosine triphosphate (ATP)
 α -motor neuron
concentric contraction
dynamic contraction
eccentric contraction
endomysium
epimysium
excitation–contraction coupling
fascicle
force–velocity relation
length–tension relation
motor unit
muscle fiber
musculoskeletal system
myofibril
myosin
myosin cross-bridge
nebulin
perimysium
plasmalemma
power stroke
principle of orderly recruitment
rate coding
sarcolemma
sarcomere
sarcoplasm
sarcoplasmic reticulum (SR)

satellite cells
single-fiber contractile velocity (V_o)
size principle
sliding filament theory
static (isometric) muscle contraction
summation
tetanus
titin
transverse tubules (T-tubules)
tropomyosin
troponin
twitch
type I fiber
type II fiber

STUDY QUESTIONS

1. List and describe the anatomical components that make up a muscle fiber.
2. List the components of a motor unit.
3. What are the steps in excitation–contraction coupling?
4. What is the role of calcium in muscle contraction?
5. Describe the sliding filament theory. How do muscle fibers shorten?
6. What are the basic characteristics that differ between type I and type II muscle fibers?
7. What is the role of genetics in determining the proportions of muscle fiber types and the potential for success in selected activities?
8. Describe the relation between muscle force development and the recruitment of type I and type II motor units.
9. Explain, and give examples of, how concentric, static, and eccentric contractions differ.
10. What two mechanisms are used by the body to increase force production in a single muscle?
11. What is the optimal length of a muscle for maximal force development?
12. What is the relation between maximal force development and the speed of shortening (concentric) and lengthening (eccentric) contractions?
13. In muscle contraction, what roles are played by the protein titin?
14. Why do previously trained muscles adapt more quickly to retraining after a period of disuse?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

